



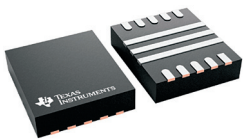
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Industry's first high-speed, single-chip lidar laser driver

TI has introduced LMH13000, the first integrated high-speed laser driver for lidar systems. It delivers an ultra-fast 800ps rise time, enabling up to 30% longer distance detection compared to traditional solutions. It also cuts



system costs by 30% and reduces module size by 75%. The lidar supports multiple signal types and eliminates the need for extra components, making it easier to integrate compact, cost-effective lidar systems into vehicles. The LMH13000 also keeps current output stable, within 2% variation, across a wide temperature range, helping systems meet Class 1 FDA eye safety standards even at higher current outputs.

Texans Instruments (TI)
<https://www.ti.com>

Mobile clock generator with built-in MEMS

SiTime Corporation has introduced Symphonic SiT30100, its first mobile clock generator with a built-in MEMS resonator. Designed for use in smartphones, tablets, laptops, and IoT devices like asset trackers, the SiT30100 delivers accurate



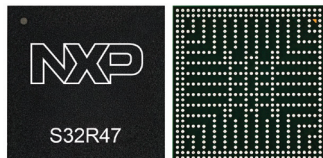
rate and reliable clock signals for 5G and GNSS chipsets while improving power efficiency. It combines four timing functions into one small chip,

helping designers save board space and simplify system design. It's built-in temperature sensor works with compensation algorithms to maintain precise timing, even under temperature changes or mechanical stress. This helps improve GPS accuracy and system performance in real-world conditions.

SiTime Corporation
<https://www.sitime.com>

Third-generation 16nm FinFET imaging radar processors

NXP Semiconductors has introduced its third-generation S32R47 imaging radar processors, built on 16nm FinFET technology. These processors deliver



up to twice the performance of the previous generation, while improving power use and reducing system cost. Combined with the company's mmWave radar transceivers, power management, and in-vehicle networking solutions, the processor meets ASIL B(D) functional safety standards (ISO 26262), making it suitable for driver-assistance systems and autonomous vehicles. The S32R47 can process three times more antenna channels in real time than current production solutions. It supports higher radar image resolution, sensitivity, and dynamic range—needed for detecting and classifying objects in varied environments such as streets or highways.

NXP Semiconductors
<https://www.nxp.com>

AI co-processor built to work with any NPU

Cadence has launched the Tensilica NeuroEdge 130 AI co-processor (AICP), a processor built to work with any neural processing unit (NPU). It supports the execution of agentic and physical AI networks used in automotive, consumer, industrial, and mobile



system-on-chips (SoCs). The AICP provides over 30% area savings and more than

20% reduction in dynamic power and energy, with no loss in performance compared to earlier versions. The AICP is made to handle tasks that are not managed by NPUs, while saving space and power. It uses the same software, AI compilers, libraries, and frameworks as other Tensilica solutions, allowing companies to bring products to market with less delay. Cadence reports customer interest and ongoing engagements.

Cadence
<https://www.cadence.com>

Automotive-grade modules with high power density

Vicor's automotive-grade power modules offer high power density and are designed to support automotive OEM and tier-one production.. The BCM6135 is a 98% efficient



2.5kW BCM bus converter that steps down 800V from the traction battery to 48V to supply safety extra-low voltage